



RV Educational Institutions
RV College of Engineering

Autonomous
Institution Affiliated
to Visvesvaraya
Technological
University, Belagavi

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New Delhi

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Academic year 2021-2022 (Even Sem)

DEPARTMENT OF
INFORMATION SCIENCE & ENGINEERING

Date	May 2022	Maximum Marks	50	
Course Code	18IS6C2	Duration	120 Min	
Sem	VI Semester	CIE-1		
ADVANCED ALGORITHMS (Common to CS & IS)				
Part – A				
Sl. No.	Questions	M	BT	CO
1.1	If $T(n) = T(n/2) + n$ for $n > 1, T(n) = 1$, then the complexity will be _____.	1	L2	CO1
1.2	In a competition, four different functions are observed. All the functions use a single for loop and within the for loop, same set of statements are executed. Consider the following for loops: a) for($i = 0; i < n; i++$) b) for($i = 0; i < n; i += 2$) c) for($i = 1; i < n; i *= 2$) d) for($i = n; i > -1; i /= 2$) If n is the size of input (positive), which function is most efficient?	2	L2	CO1
1.3.	Given an unsorted array. The array has this property that every element in array is at most k distance from its position in sorted array where k is a positive integer smaller than size of array. Which sorting algorithm can be easily modified for sorting this array and what is the obtainable time complexity?	2	L1	CO4
1.4.	What is time complexity of the following module? int fun(int n) { int count = 0; for (int i = n; i > 0; i /= 2) for (int j = 0; j < i; j++) count += 1; return count; }	2	L2	CO2
1.5	Arrange the following in the increasing order of growth $2n, \log n, n!, n, n^2, 1$.	1	L1	CO2
1.6	Bellman Ford algorithm uses which algorithm design technique _____	1	L1	CO1
1.7	_____ method of amortized analysis keeps the same amortized cost for all the operations.	1	L1	CO1
Part – B				
2a.	Define and explain the various asymptotic notations with related graphs and examples	6	L1	CO1
2b.	Solve the following recurrence relation to give the asymptotic notation using substitution method: i) $T(n) = 4T(n/2) + n^2$	4	L3	CO1
3a.	Use a recursion tree to determine a big Oh notation on the recurrence relation $T(n) = 3T(n/4) + n^2$	6	L3	CO3



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3b.	Illustrate the aggregate analysis of amortized analysis on the operation Increment in a binary counter	4	L2	CO4
4a.	State the master theorem and Mention the reason for applying and not applying the Master's theorem for the following recurrence relation i) $T(n)=9T(n/3)+n$ ii) $T(n)=T(2n/3)+1$ iii) $T(n)=2T(n/2)+2^n$ iv) $T(n)=2T(n/3)+\sin(n)$	10	L3	CO3
5a.	Using bellman Ford algorithm, find the shortest path from source vertex 'A', to the remaining vertices in the graph given below. Also mention at least two real world applications where Bellman Ford is applied.	10	L4	CO4
6a.	Find the shortest path from the source vertex to the remaining vertices in the DAG of the given graph. Use DAG shortest path algorithm. Also mention at least two real world applications where DAG shortest path algorithm is used.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test + Quiz	Max Marks	15	3	16	26	11	9	20	20	-	-

Course Outcomes:

1. Analyze various algorithms for their time and space complexity
2. Demonstrate a familiarity with major algorithms and data structures
3. Apply appropriate design techniques for solving real world problems.
4. Design and implement solutions using appropriate mathematical techniques



DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	20MAY 2022	Maximum Marks	50+10
Course Code	18CS63	Duration	110 Min
Sem	VI Semester	CIE-I	
COMPILER DESIGN			

PART A

Sl. No.	Questions	M	BT	CO
1.1	Differentiate between an interpreter and a compiler.	1	1	1
1.2	Identify the phase of the compiler which identifies the following error in the source code: <code>int a=5.0;</code>	1	2	2
1.3	_____ holds the information about variables throughout the compilation process.	1	1	2
1.4	How lexical analyzer can be divided into cascade of two processes? What is the task of each process?	2	2	3
1.5	Consider the string "RVCE" as example write the: a) substring b) proper prefix c) subsequence d) proper suffix	2	2	1
1.6	Draw a transition diagram to recognize an unsigned number.	1	2	2
1.7	Construct a leftmost derivation tree for the string " <code>((a,a),a,(a))</code> " using the grammar: $S \rightarrow (L) \mid a$ $L \rightarrow L, S \mid S$	2	3	3

PART B

Sl. No.	Questions	M	BT	CO
2.	With a neat diagram explain the structure of a compiler and show the input and output for each of the phases taking the following example statement: <code>a=5*a*(6-d)</code>	10	L1	1



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	Note: all variables are of float types			
3.a.	Define token, lexeme and pattern and identify the same in the following C statement: <code>int a; for(a=1;a<10;a++) printf("RVCE"); printf(" The value of a is %d",a);</code>	5	L2	2
3.b.	What is a "Handle"? Explain the process of handle pruning for the string <code>aaa*a++</code> for the grammar $S \rightarrow SS+ \mid SS* \mid a$	5	L2	2
4.	For each of the following grammar: a) $S \rightarrow SAT \mid TB \mid s$ $T \rightarrow aTb \mid ab$ $A \rightarrow Aa \mid a$ $B \rightarrow Bb \mid b$ b) $S \rightarrow ASA \mid aB$ $A \rightarrow BA \mid BS \mid a$ $B \rightarrow b \mid e$ Perform the following a) Modify the following grammar to suit the requirements of Top Down parser b) Find the FIRST and FOLLOW set for the following grammar	10	L3	3
5.	Write the rules for LL(1) grammar and show that the following grammar is LL(1) by constructing the Top down parsing table: $S \rightarrow AaAb \mid BbBa$ $A \rightarrow e$ $B \rightarrow e$	10	L3	3
6.	Consider the grammar given below: $E \rightarrow T + E \mid T$ $T \rightarrow \text{float} \mid \text{float} * T \mid (E)$ a) Make necessary changes to make it suitable for LL(1) parsing. b) Construct FIRST and FOLLOW sets. c) Construct the predictive parsing table. d) Show the moves made by the predictive parser on the input <code>)float**float+.</code>	10	L4	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	13	13	24	10	12	15	22	10		



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Academic Year 2021-2022 (Even Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

(Common to BT, CS, EC, EEE, ET & ISE)

Date	19/05/2022	Maximum Marks	10 + 50
Course Code	18HEM61	Duration	20 + 90 Min
Sem	VI Semester	CIE - I	
INTRODUCTION TO MANAGEMENT AND ECONOMICS			

Note: Answer all the Questions.

PART - A				
Q. No.1	Questions	M	BT	CO
1.1	List the management roles & skills and link them to management levels.	2	L1	CO1
1.2	Mention four contributions of Taylor's scientific management theory.	2	L1	CO1
1.3	Differentiate between unity of command and unity of direction.	2	L1	CO1
1.4	Write the pictorial representation of circular flow of income.	2	L1	CO4
1.5	Differentiate between price elasticity and inelasticity.	2	L1	CO4

PART - B				
Q. No.	Questions	M	BT	CO
2.	Relations theory arose from early attempts to systematically discover the psychological and social factors that lead to workplace harmony. Elucidate.	10	L2	CO1
3.	Fayol insisted that management was a skill like any other one that could be taught once its underlying principles were understood. Elaborate.	10	L3	CO1
4.	An organization should be viewed as a unified, purposeful system composed of interrelated parts. Illustrate using food industry as an example.	10	L4	CO1
5.	Examine the concept of oligopoly between two firms using a scenario of prisoner's dilemma.	10	L5	CO4
6.	Exhibit the outcomes of market demand in relation to the entry and exit of monopolistic competitors with a graphical representation.	10	L6	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	06	--	--	04	10	--	--	--	--	--
	Test	Max Marks	30	--	--	20	--	10	10	10	10	10



R V College of Engineering
Department of Computer Science and Engineering
CIE 1: Question Paper

Subject : (Code)	Artificial Intelligence & Machine Learning (18CS62)	Semester: 6th BE
Date : 15/2022	Duration : 120 minutes	Staff: Dr Sindhu/ RESA/PH/Prof Merin
Name :	USN :	Section : A/B/C/ISE

Q. No	Part-A	Marks	Level	CO
1	Give difference between Informed and Uninformed search strategies.	1	2	2
2	Task environment of agent consists of _____, _____ and _____.	1	1	1
3	Give the BFS and DFS tree traversal for the following tree. Assuming A is the start node and F is the goal node. <pre> graph TD A((A)) --- B((B)) A --- C((C)) B --- D((D)) B --- E((E)) C --- F((F)) C --- G((G)) D --- H((H)) D --- I((I)) E --- J((J)) H --- L((L)) I --- M((M)) F --- Goal(()) </pre>	2	3	2
4	What is the purpose of agent function?	1	1	1
5	Write the PEAS description of the Part Picking Robot.	2	3	1
6	Differentiate between Fully observable and partially observable environments with example.	2	2	1
7	List the parameters on which the algorithm's performance is measured.	1	1	1
Q. No	Answer all questions	Marks	Level	CO
1a	Illustrate by considering an example, learning from direct training feedback is easier than learning from indirect training feedback.	5	2	3
1b	With the example of a vacuum cleaning agent discuss the components of problem formulation.	5	2	1

2	Given an initial state of a 8-puzzle problem and final state to be reached. Find the most cost-effective path to reach the final state from initial state using A* Algorithm showing the state space clearly for each level. Consider $g(n)$ = Depth of node and $h(n)$ = Number of misplaced tiles. <table><tr><td>2</td><td>8</td><td>3</td></tr><tr><td>1</td><td>6</td><td>4</td></tr><tr><td>7</td><td></td><td>5</td></tr></table> Initial State <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>8</td><td></td><td>4</td></tr><tr><td>7</td><td>6</td><td>5</td></tr></table> Final State	2	8	3	1	6	4	7		5	1	2	3	8		4	7	6	5	10	4	2
2	8	3																				
1	6	4																				
7		5																				
1	2	3																				
8		4																				
7	6	5																				
3	Mention the types of environments suitable for the following tasks and justify your selection. i. Agent monitoring the temperature of the boiler in a manufacturing plant. ii. Hotel Recommendation iii. Agent picking defect parts from the conveyer belt iv. Radiology images analysis v. Agent Playing Chess	$2m \times 5 = 10$	2	1																		
4a	With an example discuss Model-based reflex agent	6	1,2	1																		
4b	A learning problem can be defined using Task (T), Performance measure (P), and Training Experience (E). Define the following learning problems using T, P, and E.	4	3	3																		
	i. Face recognition problem ii. To filter emails as spam or not																					
5a.	For the Graph given below apply uniform cost search and find the path from node S to G by clearly showing the states and possible paths. Discuss the algorithms performance.	6	3	2																		
5b.	Illustrate the role of Performance system, Critic, Generalizer and Experiment Generator in the final design of learning system.	4	3	3																		

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	28	19	13		6	26	18	10		

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks